

RESEARCH PROFILE

My research lies at the intersection of statistical physics, applied mathematics, and network science. I study how network structure shapes robustness and dynamical processes on networks, multilayer networks, and higher-order networks, with a focus on percolation and critical phenomena. I develop analytical and computational methods to predict how failures and spreading processes propagate through networks, and to design interventions and control strategies. Applications include spreading dynamics in epidemiology, communication and neuroscience, as well as the robustness of infrastructure systems.

Research record: 17 peer-reviewed publications and one preprint; 772 citations; h-index 12 (Google Scholar, 13 September 2026).

ACADEMIC APPOINTMENTS

University of Granada

Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellow

Granada, Spain

Sep 2026 – Present

- Host: Prof. Miguel Ángel Muñoz
- Project: Network model for mechanism of epilepsy dynamics
- Six-month secondment visit at Amsterdam UMC with Prof. Arjan Hillebrand (2027)

The Nordic Institute for Theoretical Physics (NORDITA)

WINQ Research Fellow

Stockholm, Sweden

Sep 2023 – Aug 2026

Indiana University

Visiting AccelNet/MultiNet Fellow

Bloomington, USA

Sep 2024 – Dec 2024

- Host: Prof. Filippo Radicchi

EDUCATION

Queen Mary University of London

Ph.D. in Applied Mathematics

London, United Kingdom

Sep 2019 – Sep 2023

- Advisor: Prof. Ginestra Bianconi. Thesis: “Dynamic processes on networks and higher-order structures”

University of Chinese Academy of Sciences

B.Sc. in Physics

Beijing, China

Sep 2015 – Jul 2019

- Advisor: Prof. Pan Zhang. Thesis: “Low-rank approximation of tensor networks”
- Academic visits: KTH Royal Institute of Technology (exchange, 2018); Aston University (visiting student with Prof. David Saad, 2018)

COMPETITIVE RESEARCH FUNDING

- **2025** Marie Skłodowska-Curie Actions Postdoctoral Fellowship
- **2024** AccelNet/MultiNet Fellowship and exchange funding, \$8,000
- **2023** Isaac Newton Institute Network Support funding, £5,000 (joint award)

SELECTED HONOURS AND AWARDS

- **2026** *Physical Review E* Editors' Suggestion: "Triadic percolation on multilayer networks"
- **2023** *Nature Communications* Editors' Highlight: "The dynamic nature of percolation on networks with triadic interactions"
- **2022** Outstanding Teaching Assistant nomination, King's College London
- **2021** Media coverage of "Competition, Collaboration, and Optimization in Multiple Interacting Spreading Processes" in *Discover Magazine* and by [Los Alamos National Laboratory](#)

PEER-REVIEWED PUBLICATIONS

- [1] Aida Abiad, Alex Arenas, Agnes Backhausz, Jozsef Balogh, Christopher R. S. Banerji, Sergio Barbarossa, Ginestra Bianconi, Christian Bick, Magnus Bakke B. Botnan, Timoteo Carletti, Lucia Cavallaro, Andrea Civilini, Tina Eliassi-Rad, Xue Gong, Krystal Guo, Heather Harrington, Juergen Jost, Paul L Krapivsky, Pietro Liò, Ben MacArthur, Carolina Mattsson, Pedro Mediano, Ana Paula Millán, Raffaella Mulas, Alice Patania, Giovanni Petri, Cerene Rathilal, Ruben Jose Sanchez Garcia, Martina Scolamiero, Michael T Schaub, **Hanlin Sun**, Yu Tian, Francesco Vaccarino, and Kelin Xia. "Hypergraphs and simplicial complexes in focus: A roadmap for future research in higher-order interactions". *Journal of Physics: Complexity* 7 (2026), p. 022501. DOI: [10.1088/2632-072X/ae3c4e](#).
- [2] Minsuk Kim, Lorenzo Cirigliano, Claudio Castellano, **Hanlin Sun**, Robert Jankowski, Anna Poggialini, and Filippo Radicchi. "Shortest-path percolation on scale-free networks". *Phys. Rev. E* 113 (2026), p. 014314. DOI: [10.1103/pk8t-px35](#).
- [3] **Hanlin Sun**[†], Filippo Radicchi, and Ginestra Bianconi. "Triadic percolation on multilayer networks". *Phys. Rev. E* 113 (2026), p. 014313. DOI: [10.1103/yvtg-wnn4](#).
- [4] Xinqi Hu, Gaogao Dong, Kim Christensen, **Hanlin Sun**, Jingfang Fan, Zihao Tian, Jianxi Gao, Shlomo Havlin, Renaud Lambiotte, and Xiangyi Meng. "Unveiling the importance of nonshortest paths in quantum networks". *Science Advances* 11 (2025), eadt2404. DOI: [10.1126/sciadv.adt2404](#).
- [5] Ana P Millán, **Hanlin Sun**, Lorenzo Giambagli, Riccardo Muolo, Timoteo Carletti, Joaquín J Torres, Filippo Radicchi, Jürgen Kurths, and Ginestra Bianconi. "Topology shapes dynamics of higher-order networks". *Nature Physics* 21 (2025), pp. 353–361. DOI: [10.1038/s41567-024-02757-w](#).
- [6] Ana P Millán, **Hanlin Sun**, and Joaquín J Torres. "Spatio-temporal activity patterns induced by triadic interactions in an in silico neural medium". *Journal of Physics: Complexity* 6 (2025), p. 015017. DOI: [10.1088/2632-072X/adb5e](#).
- [7] Ana P Millán, **Hanlin Sun**^{*}, Joaquín J Torres, and Ginestra Bianconi. "Triadic percolation induces dynamical topological patterns in higher-order networks". *PNAS Nexus* 3 (2024), pgae270. DOI: [10.1093/pnasnexus/pgae270](#).
- [8] **Hanlin Sun**[†] and Ginestra Bianconi. "Higher-order triadic percolation on random hypergraphs". *Phys. Rev. E* 110 (2024), p. 064315. DOI: [10.1103/PhysRevE.110.064315](#).
- [9] **Hanlin Sun**, Rajat Kumar Panda, Roberto Verdel, Alex Rodriguez, Marcello Dalmonte, and Ginestra Bianconi. "Network science: Ising states of matter". *Phys. Rev. E* 109 (2024), p. 054305. DOI: [10.1103/PhysRevE.109.054305](#).
- [10] Rajat K Panda, Roberto Verdel, Alex Rodriguez, **Hanlin Sun**, Ginestra Bianconi, and Marcello Dalmonte. "Non-parametric learning critical behavior in Ising partition functions: PCA entropy and intrinsic dimension". *SciPost Physics Core* 6 (2023), p. 086. DOI: [10.21468/SciPostPhysCore.6.4.086](#).
- [11] **Hanlin Sun**, Filippo Radicchi, Jürgen Kurths, and Ginestra Bianconi. "The dynamic nature of percolation on networks with triadic interactions". *Nature Communications* 14 (2023), p. 1308. DOI: [10.1038/s41467-023-37019-5](#).

- [12] **Hanlin Sun**, Ivan Kryven, and Ginestra Bianconi. “Critical time-dependent branching process modelling epidemic spreading with containment measures”. *Journal of Physics A: Mathematical and Theoretical* 55 (2022), p. 224006. DOI: [10.1088/1751-8121/ac65c3](https://doi.org/10.1088/1751-8121/ac65c3).
- [13] Ginestra Bianconi, **Hanlin Sun**, Giacomo Rapisardi, and Alex Arenas. “Message-passing approach to epidemic tracing and mitigation with apps”. *Phys. Rev. Research* 3 (2021), p. L012014. DOI: [10.1103/PhysRevResearch.3.L012014](https://doi.org/10.1103/PhysRevResearch.3.L012014).
- [14] Guillaume St-Onge, **Hanlin Sun**, Antoine Allard, Laurent Hébert-Dufresne, and Ginestra Bianconi. “Universal Nonlinear Infection Kernel from Heterogeneous Exposure on Higher-Order Networks”. *Phys. Rev. Lett.* 127 (2021), p. 158301. DOI: [10.1103/PhysRevLett.127.158301](https://doi.org/10.1103/PhysRevLett.127.158301).
- [15] **Hanlin Sun** and Ginestra Bianconi. “Higher-order percolation processes on multiplex hypergraphs”. *Phys. Rev. E* 104 (2021), p. 034306. DOI: [10.1103/PhysRevE.104.034306](https://doi.org/10.1103/PhysRevE.104.034306).
- [16] **Hanlin Sun**[†], David Saad, and Andrey Y. Lokhov. “Competition, Collaboration, and Optimization in Multiple Interacting Spreading Processes”. *Phys. Rev. X* 11 (2021), p. 011048. DOI: [10.1103/PhysRevX.11.011048](https://doi.org/10.1103/PhysRevX.11.011048).
- [17] **Hanlin Sun**, Robert M. Ziff, and Ginestra Bianconi. “Renormalization group theory of percolation on pseudofractal simplicial and cell complexes”. *Phys. Rev. E* 102 (2020), p. 012308. DOI: [10.1103/PhysRevE.102.012308](https://doi.org/10.1103/PhysRevE.102.012308).

PREPRINTS

- [1] Lorenzo Cirigliano, Claudio Castellano, Minsuk Kim, Filippo Radicchi, and **Hanlin Sun**[†]. *Criticality and universality in network dismantling*. 2026. arXiv: [2608.27613](https://arxiv.org/abs/2608.27613) [[physics.soc-ph](#)].

[†] corresponding author; * equal contribution.

TEACHING EXPERIENCE

- **Guest Lecturer**, Uppsala University May 2026
Lecture series on network science, with a focus on higher-order networks
- **Guest Lecturer**, Sapienza University of Rome Sep 2025
Higher-order networks and topological deep learning
- **Guest Visiting Lecturer and Examiner**, University of Iceland Jan – Apr 2025
Independently led and examined Statistical Methods in Data Processing, a full-semester 6-ECTS master’s module offered in collaboration with NORDITA and open to senior undergraduate students (10 students; 56 classes of 40 minutes over 14 weeks)
- **Teaching Associate**, Queen Mary University of London 2019 – 2023
Courses included calculus, linear algebra, and machine learning with Python (undergraduate and master’s levels)
- **Demonstrator**, Queen Mary University of London 2019 – 2021
Complex Networks, Introduction to Machine Learning, and Electricity and Atomic Physics
- **Graduate Teaching Associate**, King’s College London 2021 – 2022
Theory of Complex Networks, Linear Algebra and Geometry II, and Calculus I

SUPERVISION AND MENTORSHIP

Current supervision

- **Shanshan Liu** 2026 – Present
Master’s student, Jiangsu University; co-supervised with Prof. Gaogao Dong
Project: Critical slowing down with two scaling laws at the critical threshold of discontinuous transitions
- **Jingyu Gao** 2026 – Present
Visiting Ph.D. student, Jiangsu University
Project: The role of group diversity in higher-order spreading dynamics

Past supervision

- **Ron Demjaha** 2024
Master's project, Uppsala University
Project: Epidemic spreading model on multiplex hypergraphs
- **Yingshan Guo** 2023
Visiting master's student, Anqing Normal University; co-supervised with Prof. Ginestra Bianconi
Project: Model of recursive simplicial complexes
- **Efe Görgüner** 2022
Visiting high-school student, Royal Grammar School Guildford; co-supervised with Prof. Ginestra Bianconi
Project: The robustness and predictability of networks

ACADEMIC LEADERSHIP AND SERVICE

Society and community leadership

- **Board Member, Network Science Society** 2024 – Present
Represent [NetPlace](#) and the interests of early-career network scientists
- **Co-founder and Executive Committee Member, NetPlace** 2021 – Present
Organise online seminars and conference events for an international community of early-career network scientists

Panel organisation at NetSci and NetSciX

- **NetSciX 2026, Auckland:** *Beyond Isolation: Getting Connected in Network Science*
- **NetSci 2025, Maastricht:** *Science as a Craft*
- **NetSciX 2025, Indore:** *Challenges and Opportunities in Network Science*
- **NetSciX 2024, Venice:** panel on scientific communication
- **NetSci 2023, Vienna:** two-day programme on academic writing and mental well-being in academia

Programme and workshop organisation

- **WINQ Programme on Complex and Quantum Systems, NORDITA** 2024
Programme Organiser for a series of four workshops on complex networks and quantum systems, including the one-week international workshop *Dynamics and Topology of Complex Network Systems*, which hosted more than 20 researchers
- **NetPlace Networks & Physics, London** 2025
Workshop Organiser for a one-day research collabothon, supported by Queen Mary University of London and the British Chapter of the Network Science Society, with funding from the Isaac Newton Institute
- **DERI Ph.D. Forum, Queen Mary University of London** 2020 – 2023
Organiser of a seminar series for doctoral researchers

Editorial and peer-review service

- **Guest Editor**, Focus Issue “Focus on Higher-Order Structures in Networks and Network Dynamical Systems”, *Journal of Physics: Complexity* (2026)
- **Guest Editor**, Special Issue “Complex Dynamical Networks and Chaotic Systems”, *Mathematics*
- **Assistant Guest Editor**, Special Issue “Models, Topology and Inference of Multilayer and Higher-Order Networks”, *Entropy* (2023)
- **Peer reviewer** for 17 journals, including *Physical Review Letters*, *Science Advances*, *Nature Physics*, *Nature Communications*, *Communications Physics*, *Transportation Research Part B: Methodological*, *IEEE Transactions on Network Science and Engineering*, *Journal of Complex Networks*, and *Physical Review E*

SELECTED INVITED TALKS AND LECTURES

- Europe-Korea Conference on Science and Technology, Toulouse Jul 2026
Invited talk: The dynamic nature of percolation on networks with triadic interactions
- TopoNet 2026, NetSci satellite, Boston Jun 2026
Keynote: The dynamic nature of percolation on networks with triadic interactions
- Workshop on Random Simplicial Complexes and Hypergraphs, Nottingham Sep 2025
Invited talk: Percolation theory on higher-order networks
- 20th Course of the School of Complexity, Erice Aug 2025
Invited plenary lecture: Percolation theory on higher-order networks
- Northwestern Institute on Complex Systems, Northwestern University Nov 2024
Invited seminar: Triadic percolation induces dynamical topological patterns in higher-order networks
- Quantitative Methods for Dynamics on Networks, Los Alamos National Laboratory Aug 2024
Invited talk: The dynamic nature of percolation on networks with triadic interactions
- Isaac Newton Institute Satellite Programme on Hypergraphs, London Jul 2024
Invited talk: The dynamic nature of percolation on networks with triadic interactions
- Additional invited seminars at KTH, the Chinese Academy of Sciences, Aston University, Queen Mary University of London, and NetPlace, 2020–2023; contributed talks at NetSci, the Conference on Complex Systems, and the APS March Meeting, 2020–2026.

TECHNICAL SKILLS AND LANGUAGES

- **Programming and scientific computing:** MATLAB, Python, Mathematica, Julia, C, L^AT_EX
- **Languages:** Chinese (native); English (professional proficiency)

REFERENCES

Prof. Ginestra Bianconi

Professor
Queen Mary University of London
E1 4NS, London, United Kingdom
g.bianconi@qmul.ac.uk

Prof. Filippo Radicchi

Professor
Luddy Center for Artificial
Intelligence
Indiana University
1015 East 11th St.
Bloomington, IN 47408, USA
f.radicchi@gmail.com

Prof. David Saad

Professor
Aston University
B4 7ET Birmingham, United
Kingdom
d.saad@aston.ac.uk